

Divyesh Sankhla

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SUMMARY

Embedded Software Engineer and Embedded Firmware Engineer with 3+ years of experience building Linux-based firmware, edge applications, streaming media systems, OTA infrastructure, device management software, and production firmware for networked camera platforms. Strong in C, C++, Python, Embedded Linux, POSIX threads, IPC, synchronization, socket programming, RTSP, RTSPS, RTP, WebRTC, ONVIF, MQTT, secure communication, systemd services, Linux debugging, and low-level hardware-facing development.

Built and improved firmware used across 5+ camera SKUs, delivered OTA update infrastructure for 1,000+ deployed devices, scaled SQLite3-backed user and entitlement systems to 1M+ records/users, enhanced 15+ ONVIF APIs, and optimized real-time streaming paths using Live555 and libdatachannel. Experienced with gdb, Valgrind, Wireshark, tcpdump, ftrace, dmesg, dynamic debug, oscilloscopes, logic analyzers, Linux serial diagnostic tools, Jenkins, Docker, Git, Jira, QEMU, gdb-multiarch, and Device Tree Compiler.

Additional background includes Linux kernel modules, character drivers, platform drivers, Device Tree, U-Boot boot flow, Zephyr RTOS, FreeRTOS, interrupt handling, operating systems, formal verification, algorithms, ROS, OpenCV, CUDA, data analysis, machine learning, and hardware-software interaction.

SKILLS

Programming Languages: C, C++, C++ STL, object-oriented C++, Embedded C, Python, Bash, Shell scripting, SQL, SQLite3, MySQL, Assembly language, Verilog, MATLAB

Embedded Linux and Platform Firmware: Embedded Linux, Linux internals, Linux kernel, Linux kernel modules, device drivers, character drivers, platform drivers, file operations, dynamic device numbers, dev node creation, ioctl, poll/select, Device Tree, Device Tree Compiler, U-Boot, bootloader flow, bootargs, kernel handoff, systemd services, daemon lifecycle management, runtime process control, service recovery, OTA firmware systems, staged downloads, A/B partition switching, rollback recovery, checksum validation, image verification, version gating, audit logging, post-update health monitoring, Ubuntu Core

RTOS, Operating Systems and Concurrency: Zephyr RTOS, FreeRTOS, Pintos, operating systems, systems programming, POSIX threads, POSIX timers, IPC, multithreading, synchronization, mutexes, semaphores, spinlocks, deferred work, work queues, message queues, memory management, process management, system calls, file I/O, exec/wait, CPU scheduling, priority scheduling, priority donation, advanced scheduler behavior, user-kernel transitions, user-kernel memory safety, interrupt handling, ISR-safe deferred processing, SMP concepts, computer architecture

Networking, Streaming and Media: TCP/IP, TCP, UDP, POSIX sockets, socket programming, client-server architecture, select I/O multiplexing, message framing, network simulation, NS-3, network performance analysis, LAN-WAN, HTTP, HTTPS, WebSockets, MQTT, Mosquitto, ONVIF, ONVIF G/S/T profiles, REST APIs, SOAP APIs, gSOAP, Lighttpd, RTSP, RTSPS, RTP, WebRTC, Live555, libdatachannel, FFmpeg, NAT, DHCP, STUN, TURN, RTP pacing, jitter buffering, adaptive buffering, secure session setup, multi-client streaming

Security and Access Control: TLS, SSL, OpenSSL, digest authentication, cryptography, symmetric encryption, public-key encryption, RBAC, entitlements, access control, feature-level permissions, secure OTA image verification, audit logging, computer security

Debugging, Validation and Tools: gdb, Valgrind, ftrace, dmesg, dynamic debug, stack traces, Wireshark, tcpdump, Postman, MQTtX, oscilloscope, logic analyzer, Linux serial diagnostic tools, packet capture, telemetry workflows, RTT analysis, packet-loss analysis, latency analysis, hardware debugging, software troubleshooting, Jenkins, Docker, Git, Jira, STM32CubeIDE, QEMU, gdb-multiarch, Device Tree Compiler

Data, Machine Learning and Analytics: Data cleaning, exploratory data analysis, feature engineering, telemetry, metrics, XGBoost, Decision Trees, SVM, MLP, Logistic Regression, K-Means clustering, Stacking, Tableau, Power BI, model evaluation, statistical modeling

Algorithms and Formal Verification: Algorithms, data structures, algorithm analysis, theorem proving, Lean4, Mathlib, formal verification, asymptotic notation, recurrence relations, Masters Theorem, divide-and-conquer algorithms, dynamic programming, greedy algorithms, Merge Sort, Count Inversions, Insertion Sort, Interval Scheduling, Decision Subset Sum, Optimal Subset Sum

Hardware, Interfaces and Platforms: UART, RS485, serial communication, packet framing, CRC validation, retry logic, buffering, microcontrollers, Internet of Things, PCB design, ROS, OpenCV, CUDA, Simulink, Arduino IDE, gem5, RISC processor design, instruction set architecture, CPU design, Ambarella CV22, Ambarella CV2, NVIDIA Jetson TX2, STM32, Raspberry Pi, ARMv7, ARMv8, 8051, 8086, FPGA, AI fixed cameras, PTZ cameras, intelligent network surveillance cameras, autonomous vehicles

Engineering Practices: Firmware, embedded firmware, embedded systems, embedded devices, embedded software programming, platform firmware, system software, systems management, IP cameras, streaming media, software requirements, product requirements, coding standards, code review, source control management, build processes, testing, operations, system development, design review, technical documentation, architecture documentation, problem solving

EXPERIENCE

University at Buffalo

Buffalo, NY

Research Assistant

Oct 2025 - Apr 2026

- Developed Lean4 formal proofs and reusable Mathlib libraries for asymptotic notation, divide-and-conquer algorithms, recurrence relations, and a general Masters Theorem framework.
- Formalized algorithm correctness for Merge Sort, Count Inversions, Insertion Sort, Interval Scheduling, Decision Subset Sum, and Optimal Subset Sum.
- Strengthened proof-based software development skills through theorem proving, functional programming, algorithm analysis, and formal verification.

eInfochips, An Arrow Company

Ahmedabad, India

Embedded Software Engineer

Aug 2021 - Dec 2024

- Developed multithreaded embedded C/C++ services for Linux IP camera platforms across 5+ SKUs, supporting streaming, recording, networking, entitlement, OTA, device management, systemd runtime control, and service recovery workflows.
- Delivered production Linux camera firmware for intelligent surveillance devices, including AI fixed and PTZ camera platforms, while working across firmware, QA, hardware, and cloud teams.
- Collaborated with diverse international teams of 50+ members through software development life cycle activities, feature integration, debugging, validation, and release support.
- Implemented ONVIF server services using gSOAP and Lighttpd with POSIX threads, IPC, mutex synchronization, daemon lifecycle handling, and request processing across Embedded Linux environments.
- Built ONVIF Server Profile and WebSocket APIs across 5+ SKUs, enhanced 15+ APIs, and supported NVR/VMS integration for intelligent network surveillance cameras.
- Scaled SQLite3-backed user and entitlement data to 1M+ records/users using safer database migration and access patterns, including SQLite3 indices and WAL for high-QPS reads.
- Delivered OTA firmware update infrastructure for 1,000+ deployed devices with staged downloads, A/B partition switching, rollback recovery, checksum validation, image verification, version gating, audit logging, and post-update health monitoring.
- Integrated OTA upgrades into Linux camera firmware while preserving data integrity and supporting smooth transitions across deployed devices.
- Designed low-latency Live555 RTSP and WebRTC streaming paths using multithreaded C++, libdatachannel, Mosquitto signaling, RTP pacing, jitter buffering, scheduler-aware buffering, and lock-aware queues.
- Optimized RTSP/WebRTC media pipelines for multi-client streaming by tuning buffering, RTP pacing, scheduler-aware processing, and secure TLS/digest authentication.
- Standardized telemetry and packet-capture workflows using RTT, packet loss, latency, tcpdump, and Wireshark to diagnose streaming issues and improve player-stack behavior.
- Engineered a Cloud Video Recorder module with RTSPS push, TLS/digest authentication, event-based recording, C++ thread/queue-based buffer management, pre/post buffers, backpressure, retry handling, event indexing, retention, and a 30-second local fallback buffer for WAN loss.
- Built MQTT-based centralized configuration management across 4+ camera SKUs with transactional rollback handling, synchronized state updates, schema validation, persistent storage, MQTT fanout, and cloud app integration.
- Implemented an Entitlement Manager with RBAC-style access control across 5+ camera functionalities and integrated entitlement checks with device/cloud workflows.
- Built access-control logic for user and feature permissions, supporting secure configuration and service behavior across multiple camera SKUs.
- Developed an event manager module that enabled cloud-based intelligent event recognition and NLP-based event search.
- Built an event-intelligence pipeline to collect, normalize, and publish camera events to cloud services for downstream search, analytics, intelligent recognition, and NLP-based search.
- Refactored large C/C++ legacy firmware modules by consolidating repositories, upgrading SQLite3 workflows, unifying JSON/XML handling with cJSON/libxml2, improving database migration behavior, and reducing redundant implementation patterns across a 7-member team.
- Refactored multi-sensor camera architecture across 3 camera series by redesigning reusable configuration workflows, sensor-indexed processing, and scalable multi-camera support.
- Debugged Linux networking and concurrency issues involving TCP CLOSE-WAIT socket leaks, race conditions, synchronization bottlenecks, and multithreaded execution using gdb, Valgrind, ftrace, tcpdump, and Wireshark.
- Managed systemd-based Linux services with runtime process control, daemon coordination, service recovery, and production debugging across embedded camera platforms.

Krossmark Innovations

Ahmedabad, India

Embedded Engineering Intern

Jun 2020 - Jul 2020

- Implemented RS485/UART communication handling with CRC validation, retry logic, buffering, packet framing, and low-level debugging using oscilloscopes, logic analyzers, and Linux serial diagnostic tools.

- Developed RS485/UART sensor communication with CRC, buffering, and retry logic, then verified packet integrity using Linux tools, oscilloscopes, and logic analyzers.

PROJECTS

Linux IP Camera Firmware Across 5+ SKUs

- Built Linux-based firmware modules for intelligent network surveillance cameras across Honeywell AI fixed/PTZ camera products and Hanwha cloud-integration camera SKUs.
- Developed embedded C/C++ applications and multithreaded Linux services for streaming, networking, recording, entitlement, OTA, device management, ONVIF compatibility, access control, cloud configuration, and firmware upgrades.
- Used POSIX threads, IPC mechanisms, synchronization, mutexes, daemon lifecycle handling, systemd, SQLite3, JSON/XML handling, cJSON, libxml2, gSOAP, Lighttpd, MQTT, Mosquitto, RTSP, RTSPS, RTP, WebRTC, Live555, libdatachannel, TLS/SSL, and digest authentication.
- Delivered firmware functionality across 5+ SKUs, supported OTA updates for 1,000+ deployed cameras, supported user/entitlement database scale up to 1M+ records/users, and improved shared firmware code for multiple camera models.

ONVIF Server APIs and WebSocket Services

- Created and enhanced ONVIF Server Profile APIs and ONVIF WebSocket APIs for HTTP request access across Embedded Linux camera platforms.
- Implemented ONVIF server services using gSOAP, Lighttpd, POSIX threads, IPC mechanisms, mutex synchronization, daemon lifecycle handling, and concurrent request processing.
- Integrated camera series with ONVIF-supported NVRs and VMS systems, enhanced 15+ APIs, improved compatibility across 5+ SKUs, and supported large-scale user and entitlement handling.
- Worked on SQLite3 database upgrade and scaling, SQLite3 indices, WAL, safer migrations, JSON/XML parser logic, and later JSON/XML consolidation.

OTA Firmware Update Infrastructure

- Integrated OTA firmware upgrade workflows into Linux camera firmware for more than 1,000 deployed/market cameras.
- Designed and deployed staged downloads, checksum validation, image verification, version gating, A/B partition switching, rollback recovery, post-update health checks, durable logs, audit records, and data-integrity safeguards.
- Worked with SQLite3 database upgrades and migration workflows as part of firmware update transitions.
- Improved safety of fleet-wide updates and reduced risk of failed updates by adding verification, rollback, health monitoring, and persistent update records.

Low-Latency RTSP, RTSPS and WebRTC Streaming Module

- Designed low-latency Live555 RTSP and WebRTC streaming paths using multithreaded C++, libdatachannel, Mosquitto signaling, RTP pacing, jitter buffering, scheduler-aware buffering, and lock-aware queues.
- Built multi-client handling and peer management using C++ classes, C++ STL queues, linked-list structures, POSIX threads, and secure session setup.
- Tuned buffering, RTP pacing, jitter behavior, adaptive buffering, and lossy-network behavior to improve global real-time IP camera streaming stability.
- Used TLS, digest authentication, NAT, STUN, TURN, FFmpeg, telemetry, metrics, tcpdump, and Wireshark to diagnose player-stack and network issues.

Cloud Video Recorder With RTSPS Push

- Engineered a Cloud Video Recorder module using C++ classes, queues, and threads to securely push video clips to cloud services.
- Implemented RTSPS push, TLS and digest authentication, event-based recording, pre/post event buffers, back-pressure, retry behavior, event indexing, retention, and buffer management.
- Added a minimum 30-second local fallback buffer for WAN/network loss, improving cloud recording reliability and reducing event loss during outages.

MQTT-Based Centralized Configuration and Settings Manager

- Built a cloud-integrated settings module using MQTT/Mosquitto to apply, validate, persist, synchronize, and roll back device configuration changes safely across multiple SKUs.
- Designed centralized configuration management across 4+ camera SKUs with MQTT fanout, persistent configuration storage, transactional apply/rollback, schema validation, synchronized state updates, and cloud app integration.
- Reduced misconfiguration risk across camera SKUs and improved resilience of device settings updates.

Entitlement Manager and Access Control

- Implemented RBAC-style entitlement logic for access control across 5+ camera functionalities.
- Connected entitlement behavior with ONVIF services, cloud-facing modules, MQTT/config integration, SQLite3-backed user/entitlement storage, and device workflows.
- Improved feature-level permissions and secure configuration behavior across multiple camera SKUs.

Event Manager and Cloud Event Intelligence

- Developed an event manager module that collected device events such as motion and analytics events from camera firmware.
- Built an event-intelligence pipeline to normalize events before publishing them to cloud services.
- Enabled cloud-based intelligent event recognition, searchable intelligent events, downstream analytics, and NLP-based event search.

Repository Refactoring, SQLite3 Upgrade and JSON/XML Consolidation

- Refactored large C/C++ legacy modules by consolidating repositories, upgrading SQLite3 workflows, and unifying JSON/XML handling across a 7-member team.
- Modified C/C++ functions and classes, standardized JSON/XML parser handling using cJSON/libxml2, improved database migration behavior, and reduced repeated implementation patterns across modules.
- Improved codebase efficiency, reduced integration risk, reduced integration defects, and made shared firmware code easier to maintain.

Multi-Sensor Camera Architecture Refactoring

- Refactored multi-sensor camera architecture across 3 camera series by redesigning reusable configuration workflows, sensor-indexed processing, and scalable multi-camera support.
- Reworked camera configuration handling around reusable workflows, reduced hard-coded camera-series logic, improved support for multiple sensors/models, and made multi-camera firmware easier to maintain and extend.

Linux Networking and Concurrency Debugging

- Debugged TCP CLOSE-WAIT socket leaks, race conditions, synchronization bottlenecks, and multithreaded service behavior in Embedded Linux camera services.
- Used gdb, Valgrind, ftrace, tcpdump, Wireshark, RTT, packet loss, latency telemetry, and packet-capture workflows to trace runtime failures and improve service stability.

RS485/UART Communication Handling

- Implemented RS485/UART communication handling for sensor/device communication with packet framing, CRC validation, retry logic, buffering, and data-integrity checks.
- Used Linux serial diagnostic tools, oscilloscopes, and logic analyzers to inspect low-level signal behavior and verify packet integrity.

Linux Kernel and Driver Development

- Built and debugged Linux kernels in QEMU with debug configuration, annotated boot logs, and kernel boot flow notes from start kernel to userspace init.
- Developed Linux kernel modules and character/platform drivers with file operations, dynamic device numbers, dev node creation, ioctl, poll/select, and userspace test applications.
- Implemented interrupt handling, deferred work, mutex/spinlock synchronization, blocking/non-blocking behavior, and Device Tree based platform-driver integration.
- Debugged kernel crashes, race conditions, and driver execution paths using dmesg, stack traces, ftrace, and dynamic debug.

Bootloader and U-Boot Bring-Up

- Built and customized U-Boot from source in QEMU, booted Linux through U-Boot, and documented reset-to-kernel handoff.
- Implemented a custom U-Boot command, modified boot arguments, manually loaded kernel, Device Tree, and initrd images, and debugged early boot failures.
- Studied U-Boot memory layout using bdinfo, stack/heap relocation, Device Tree properties, and kernel handoff behavior.

Zephyr RTOS Firmware and Driver Development

- Developed Zephyr RTOS applications with multiple threads, priorities, scheduling analysis, logging, and shell support.
- Implemented custom GPIO/I2C driver structure using DEVICE DT DEFINE, Device Tree bindings, driver API structs, and sample applications.
- Designed ISR-safe interrupt handling with deferred processing through work queues, message queues, or semaphores, measured timing behavior, and built power-aware firmware with sleep/idle states and interrupt/timer wakeup.

Pintos Operating System Kernel

- Implemented alarm clock, priority scheduling, priority donation, advanced scheduler behavior, recent CPU, load average, timer-driven preemption, synchronization primitives, system calls, file I/O, exec/wait, and safe user-kernel transitions in the Pintos OS kernel using C.
- Built memory-safety checks and debugged low-level concurrency, synchronization, and memory-protection behavior in kernel C.

Socket-Based Chat Application

- Built a TCP-based client-server chat application in C using POSIX sockets with login, direct messaging, broadcast, blocking, offline buffering/storage, message framing, select I/O multiplexing, and CLI command handling.
- Implemented a single-threaded socket server, client state machines, framed messages, and real-time chat workflows.

Autonomous Underwater Vehicle

- Led development of an Autonomous Underwater Vehicle using NVIDIA Jetson TX2, ROS, OpenCV, CUDA, sensors, computer vision, PID control, thruster control, and real-time data exchange.
- Built a ROS publish/subscribe graph for communication between perception and control nodes, integrated sensor data with navigation logic, and built a computer-vision/control pipeline for underwater path planning.
- Implemented color detection, shape detection, contouring, PID loops for thruster control, precise underwater navigation, and CUDA acceleration for vision/perception performance.
- Achieved 15x CUDA speedup in one project version and improved real-time path planning and maneuverability.

Digital Well-Being Data Analysis and Machine Learning

- Analyzed 300K+ social media/digital wellness records using Python, data cleaning, exploratory data analysis, feature engineering, XGBoost, SVM, MLP, Logistic Regression, Decision Tree, Stacking, and K-Means.
- Modeled outcomes such as screen time, sleep quality, social media fatigue, loneliness, health indicators, and physical exercise effects.
- Built ML/statistical models and identified relationships between prolonged screen use, poor health indicators, and the mitigating role of physical exercise.

EDUCATION

University at Buffalo, The State University of New York

Master of Science in Computer Science and Engineering

Buffalo, NY

Jan 2025 - May 2026

Nirma University / Institute of Technology, Nirma University

Bachelor of Technology in Electronics and Communication Engineering

CPI: 7.31/10

Ahmedabad, India

Jul 2017 - Jun 2021

PUBLICATIONS

- Industrial Automation Using PLC, and IoT Advancements, IJERT, manuscript accepted, May 2021.

ACHIEVEMENTS

- Power Trainee Award, eInfochips Annual Award Ceremony, 2022.
- Best Team Award, eInfochips Annual Award Ceremony, 2024.
- Pat on the Back Award, eInfochips / Duclo.

VOLUNTEER EXPERIENCE

- Assisted with administrative tasks, facility upkeep, and staff support for resident services at an old age home.
- Taught coding and digital literacy to underprivileged students through an NGO.
- Organized a campaign and distributed food rations and clothing to people in need.